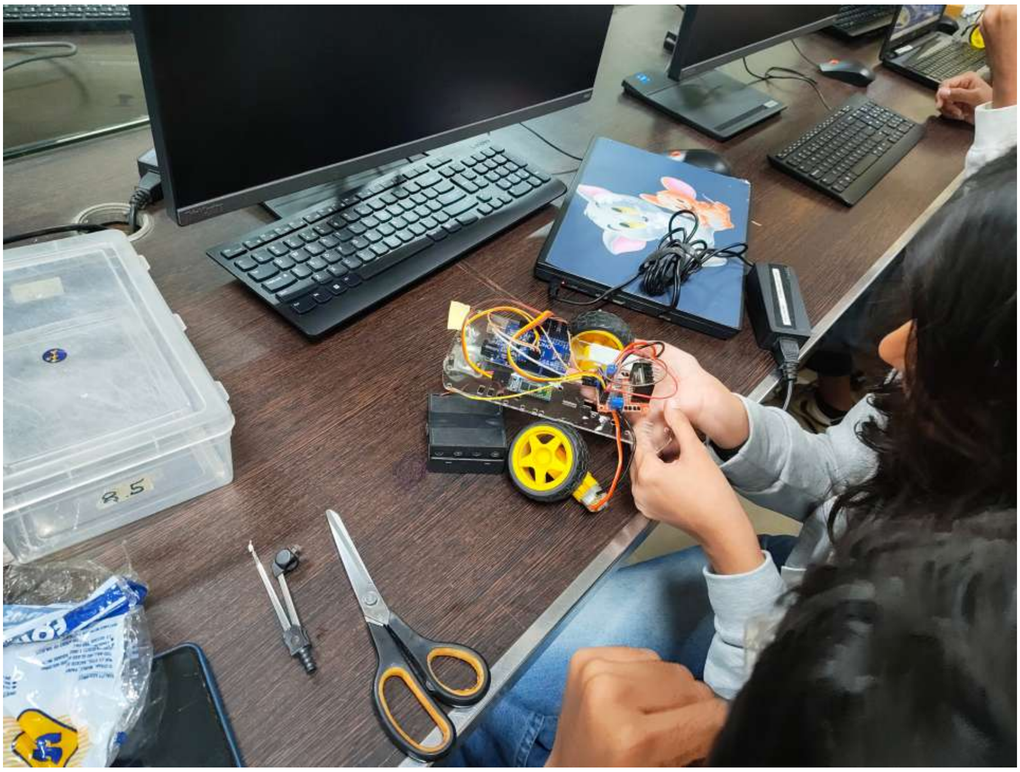
Android Controlled Car Using Arduino and Bluetooth Technology

Project Report

Domain	Embedded Systems | Internet of Things (IoT) | Robotics
Project Type	Mini Project
Project Duration	January 2026 - February 2026
Presented By	Vedant Modi
Event	Technovation'26 - Usha Pravin Gandhi College of Arts, Science and Commerce



The assembled Android Controlled Car

Project Overview

The Android Controlled Car is a wireless robotic vehicle developed using the Arduino Uno microcontroller and Bluetooth communication technology. The system enables users to control the movement of the vehicle through an Android smartphone using a Bluetooth controller application.

The project combines embedded systems, electronics, robotics, and mobile communication to demonstrate how wireless technology can be used for remote vehicle control. Commands sent from the Android device are transmitted through the HC-05 Bluetooth module to the Arduino, which processes the instructions and controls the motors through an L298N motor driver.

The project was developed to gain practical knowledge of Arduino programming, motor control, embedded system design, and wireless communication. It was successfully presented during Technovation'26, where it received a Certificate of Participation for successful development and presentation.

Problem Statement

Remote-controlled robotic vehicles are widely used in industries, research laboratories, surveillance, education, and automation. Traditional wired robots suffer from limited mobility and restricted operating range.

The objective of this project was to eliminate wired control by implementing a Bluetooth-based wireless communication system. This allows users to remotely operate the vehicle through an Android smartphone, making the system more flexible, portable, and user-friendly.

Objectives

- To design and develop a wireless robotic car using Arduino.
- To establish Bluetooth communication between an Android smartphone and the robot.
- To understand the fundamentals of embedded systems and robotics.
- To implement motor control using an L298N motor driver.
- To gain hands-on experience with Arduino programming and electronic circuit design.
- To demonstrate practical applications of IoT and wireless communication technologies.

Hardware Components

Component	Purpose
Arduino Uno	Main microcontroller
HC-05 Bluetooth Module	Wireless communication
L298N Motor Driver	Controls DC motors
DC Motors	Provides movement
Robot Chassis	Structural body

Wheels	Movement
Rechargeable Battery Pack	Power supply
Jumper Wires	Electrical connections
Switch	Power control

Software Requirements

- Arduino IDE
- Embedded C / Arduino Programming Language
- Bluetooth Controller Android Application
- Windows Operating System

Working Principle

The Android Controlled Car works by establishing Bluetooth communication between an Android smartphone and the HC-05 Bluetooth module connected to the Arduino Uno.

After pairing the smartphone with the Bluetooth module, the user opens the Bluetooth Controller application. Whenever the user presses a movement button, the application sends a specific character command through Bluetooth.

The HC-05 module receives the command and forwards it to the Arduino Uno through serial communication. The Arduino interprets the received instruction and activates the appropriate outputs connected to the L298N motor driver.

The motor driver controls the rotation of the DC motors, allowing the robot to move forward, backward, left, right, or stop depending on the received command.

This communication process occurs in real time, enabling smooth and responsive control of the robotic vehicle.

System Workflow

Android Smartphone

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Bluetooth Commands

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HC-05 Bluetooth Module

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Arduino Uno

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L298N Motor Driver

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DC Motors

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Robot Movement

Features

- Wireless control using Bluetooth.
- Android smartphone compatibility.
- Real-time command processing.
- Forward movement.
- Reverse movement.
- Left and right turning.
- Stop functionality.
- Rechargeable battery-powered operation.
- Compact and lightweight design.
- Easy hardware maintenance.
- Low-cost implementation.
- Beginner-friendly embedded systems project.

Development Methodology

The project was completed in multiple stages.

Initially, all hardware components including the Arduino Uno, motor driver, Bluetooth module, motors, battery pack, and robot chassis were assembled.

After hardware assembly, the Bluetooth module was connected to the Arduino, while the motor driver was interfaced with both the Arduino and the DC motors.

Arduino IDE was then used to write and upload the control program into the Arduino Uno. The program was designed to receive Bluetooth commands and control motor movement accordingly.

After successful programming, the Bluetooth module was paired with an Android smartphone using a Bluetooth Controller application. Multiple test runs were conducted to verify communication stability and movement accuracy.

Finally, the complete system was demonstrated successfully during Technovation'26.

Challenges Faced

Several technical challenges were encountered during project development.

One of the major challenges was establishing stable Bluetooth communication between the Android application and the HC-05 module. Proper pairing and baud rate configuration were essential for reliable communication.

Motor wiring also required careful attention because incorrect polarity resulted in unexpected movement directions.

Battery voltage management was another challenge, as low battery levels affected motor performance.

Debugging both hardware connections and Arduino code required multiple iterations before achieving stable operation.

Despite these challenges, the project was successfully completed and demonstrated.

Applications

The Android Controlled Car has numerous real-world applications.

It can be used for educational demonstrations to teach robotics and embedded systems concepts. Similar technology is widely used in warehouse automation, industrial material transportation, military reconnaissance, surveillance systems, hazardous environment inspection, rescue operations, and smart robotics research.

The project also serves as an excellent foundation for advanced IoT-based robotic systems.

Skills Acquired

- Arduino Programming
- Embedded Systems Development

- Internet of Things (IoT)
- Robotics
- Bluetooth Communication
- Electronic Circuit Design
- Motor Driver Integration
- Hardware Troubleshooting
- Sensor Interfacing
- Debugging
- Problem Solving
- Team Collaboration
- Technical Documentation

Project Outcomes

The Android Controlled Car was successfully designed, assembled, programmed, and tested. The project demonstrated reliable Bluetooth communication between an Android smartphone and the robotic vehicle.

It strengthened practical knowledge in embedded systems, robotics, electronic hardware, and Arduino programming while improving troubleshooting and teamwork skills.

The project was successfully showcased at Technovation'26, earning a Certificate of Participation for its successful development and presentation.

Future Scope

The project can be further enhanced by integrating modern technologies such as:

- Wi-Fi Control using ESP32
- Voice Control using Google Assistant
- Obstacle Avoidance using Ultrasonic Sensor
- Line Following Capability
- GPS Tracking
- Live Camera Streaming
- Mobile Application Development
- Autonomous Navigation
- Artificial Intelligence Integration
- Cloud-Based IoT Monitoring

Conclusion

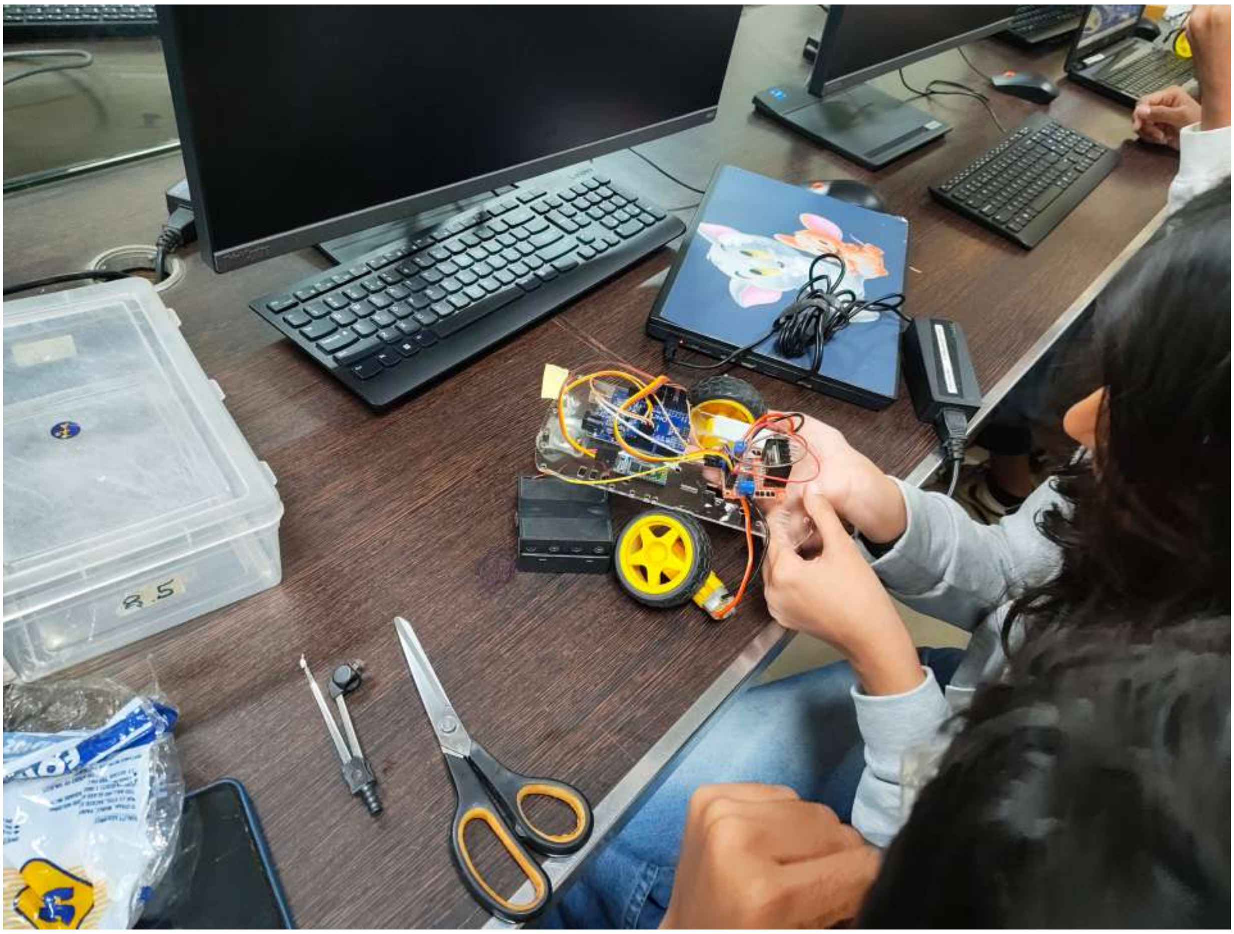
The Android Controlled Car project successfully demonstrated the implementation of embedded systems and wireless communication using Arduino and Bluetooth technology. The project achieved its objectives by enabling users to remotely control a robotic vehicle through an Android smartphone.

This project provided valuable practical experience in electronics, robotics, Arduino programming, and IoT while enhancing problem-solving and hardware integration skills. It also showcased the importance of embedded systems in modern automation and served as an excellent learning experience in developing real-world engineering solutions.

Project Evidence

Project Photograph

Photograph of the assembled Android Controlled Car:



Certificate of Participation

Technovation'26 Certificate of Participation, certifying the successful development and presentation of the project titled “Android Controlled Car”, awarded to Vedant Modi.

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CERTIFICATE OF PARTICIPATION

PRESENTED TO

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This certificate certifies the successful development and presentation of the project Titled
Android Controlled Car

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